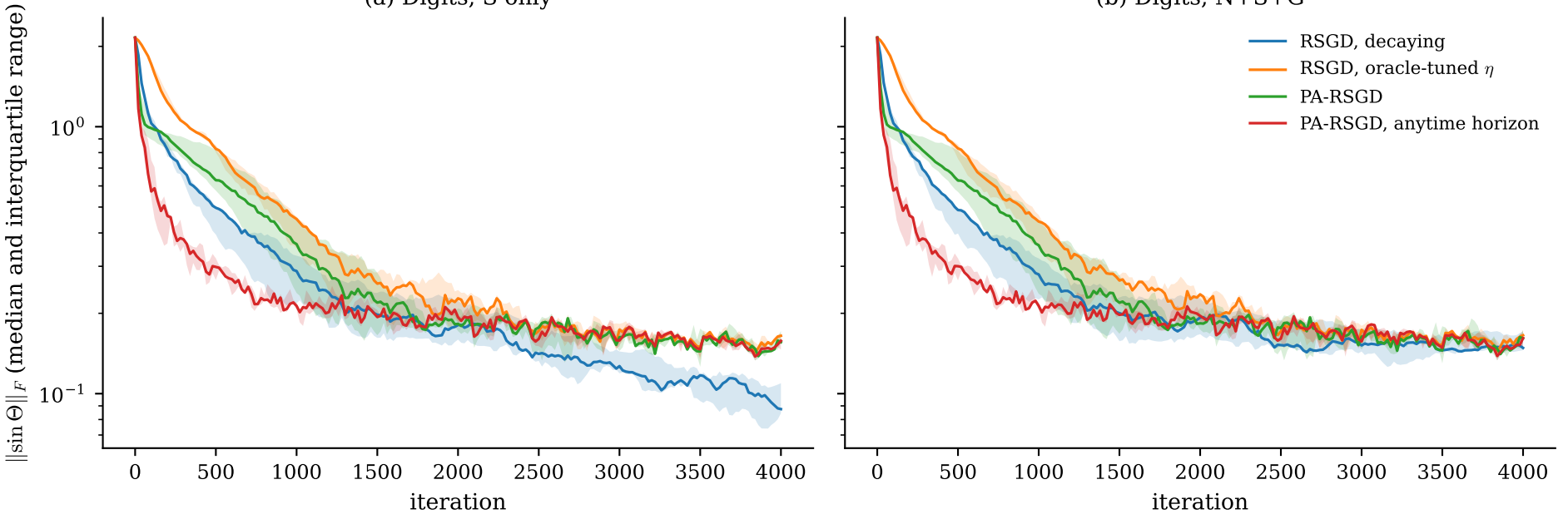

(a) Digits, S only
(b) Digits, N+S+G
$\|\sin\Theta\|_F$ (median and interquartile range)
$10^0$
$10^{-1}$
0
500
1000
1500
2000
2500
3000
3500
4000
iteration
RSGD, decaying
RSGD, oracle-tuned $\eta$
PA-RSGD
PA-RSGD, anytime horizon